

Joshua Greenhalgh

Senior Backend Software Engineer

United Kingdom

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Summary

Backend software engineer working in Python and PostgreSQL/PostGIS since 2017, in senior roles across biotech, adtech, climate analytics and humanitarian data, including two years at the non-profit Flowminder Foundation. Builds the backend systems that ingest, validate and transform large partner and customer datasets: geospatial workflows on PostGIS, satellite imagery pipelines feeding vision models, data checks, error handling and observability, and Terraform-managed containerised services with CI/CD on AWS and GCP. Traces problems through data and system behaviour, is comfortable with ambiguous and evolving requirements, mentors engineers and explains complex ideas clearly to non-technical colleagues, backed by a first-class mathematics degree and doctoral research in computational modelling.

Professional Experience

Independent Software & ML Projects

Open-source tooling and experiments, UK, May 2026 – Present

- Building and open-sourcing ML tooling: an activation-probing toolkit and a weight-perturbation experiment framework for open language models, a Modal and Weights & Biases experiment runner, and an LLM-based knowledge-graph extraction pipeline (see Projects).

Senior Software Engineer

IMU Biosciences, UK, Feb 2024 – May 2026

- Designed and built the **data platform and catalogue** for 100k+ single-cell (flow cytometry) samples, unifying scattered Parquet and vendor file sources into a structured, lineage-aware index. Cut query times and made ML training datasets fully reproducible.
- Owned **storage architecture** for high-volume single-cell data end to end: benchmarked Zarr and Parquet access patterns across S3 and Snowflake and delivered a hybrid design serving both whole-sample retrieval and cross-sample aggregation for model training.
- Built a **QC and monitoring framework** in Dagster that automatically validates reference-sample consistency across experiments, catching data-quality regressions before they reach analysts or models.
- Defined **domain ontologies and metadata standards** with immunologists and embedded them in pipeline outputs, standardising labelling across diverse assays and improving the interpretability of downstream ML models.
- Developed a **synthetic data generation framework** that produces realistic single-cell datasets, used to stress-test pipelines and validate ML model generalisation.

Career break: planned time off between roles, walked the Camino de Santiago (Oct 2023 – Jan 2024)

Senior Data Engineer

Mobkoi, UK, Jan 2022 – Oct 2023

- Rebuilt the company's data pipelines from **ad-hoc cloud functions to Prefect orchestration** on GCP with CI/CD, increasing deployment speed and safety while reducing operational overhead.
- Led migration from aggregated partner data to **impression-level ingestion**, integrating with AWS ad-serving infrastructure and CloudWatch logs, enabling granular analytics and increasing client confidence in reported numbers.
- Designed and implemented a **new data warehouse** with change data capture from internal operational systems, improving data freshness and reliability for the analytics team.
- Overhauled **revenue estimation and FX handling**, reducing monthly variance against P&L from ~20% to under 1%.

Co-founder

Personal location data union (co-founded with a former Flowminder colleague), UK, Apr 2021 – Dec 2021

- Two-sided data venture: individuals share their location data on their own terms and are paid for it, and buyers receive **privacy-preserving mobility aggregates**, building on Flowminder experience of turning raw location data into aggregate mobility products.
- Built a proof-of-concept **mobile app (Expo/React Native) and location collection backend on GCP** (BigQuery, Terraform) end to end, from consent and data capture through storage and aggregation. Discontinued after finding the pay-for-data model commercially unviable.

Senior Data Engineer

TransitionZero, UK, Jan 2021 – Mar 2021

- Implemented **Terraform-managed infrastructure on GCP** with revision-based deployments, and replaced Cloud Composer with **Prefect pipelines** and a CI-based framework that let analysts deploy their own pipelines independently.
- Built satellite imagery processing pipelines on **Google Earth Engine** feeding vision-based ML models for climate analytics.

Senior Data Engineer

HeyJobs, Berlin, Germany, Aug 2019 – Nov 2020

- Built a **data quality framework for the Redshift warehouse** using property-based testing of SQL transformations, integrated with Airflow, DataDog and Slack as part of CI. Significantly reduced post-deployment data issues.
- **Tech lead** for redesigning marketing cost processing, migrating undocumented legacy SQL to fully unit-tested Python and reducing marketing overspend by ~5%.
- Established **Airflow best practices** (incremental, idempotent pipelines) and moved intermediary data to Glue/Spectrum, cutting Redshift costs.

Data Engineer / Analyst

Flowminder Foundation, Southampton, UK & Berlin (remote), Jun 2017 – Jun 2019

- **Led a GSMA-funded project end to end**, including budget and contractors, standardising ETL on **Airflow** across multiple countries (50M+ mobile phone records ingested daily) and building a post-crisis mobility analysis tool on Kafka and TimescaleDB for humanitarian aid targeting.
- Led an **Inter-American Development Bank** project estimating commuting flows in Haiti's two largest cities from three years of mobile phone data (~10M subscribers), delivering interactive PostGIS and React dashboards to the transport ministry.
- Designed and delivered a two-week **Python and R training programme** for Nepal's National Statistics Office on JupyterHub (GKE).

Doctoral Researcher, Computational Modelling

University of Southampton, Southampton, UK, Sep 2015 – Jun 2017

- Researched **deep learning, optimisation and Bayesian methods** for inverse problems in x-ray tomography (TensorFlow, PyMC3, OpenMP/MPI on HPC). Taught and mentored undergraduates in Python and mathematics and postgraduates in research software engineering. Left to accept a permanent offer from Flowminder after an internship there.

Projects

Selected personal ML & LLM projects, open source at github.com/josh-gree (2025 – 2026)

- **probe-lab**: toolkit for LLM activation probing. Extracts per-layer residual-stream activations with PyTorch and Hugging Face Transformers and fits linear probes against prompt metadata for interpretability experiments.
- **weight-corruption**: experimental framework measuring how weight perturbations (noise, zeroing, shuffling, quantisation) degrade language modelling and behaviour across the Pythia model family, with GPU evaluation on Modal.
- **birdclef-2026** and **wm**: bird-audio classification in PyTorch (timm, ONNX embeddings) trained on Modal GPUs with Weights & Biases tracking; wrote **wm**, a small framework that snapshots every experiment run as a PR for reproducibility.
- **news-knowledge-graph**: LLM extraction pipeline turning news articles into a knowledge graph using DSPy programs and GLiNER2 for entity extraction, temporal tagging and Wikidata entity linking, orchestrated with Prefect.

- **mobile-agents:** GitHub Actions workflow built on the Claude Agent SDK that plans and implements issues autonomously and opens pull requests.

Skills

Languages & Backend: Python, SQL, Bash, Django, Flask, REST APIs, PySpark, R, JavaScript

PostgreSQL: PostgreSQL, PostGIS, TimescaleDB, data modelling, joins, query optimisation, performance tuning

Geospatial: PostGIS, Google Earth Engine, satellite imagery pipelines, mobility aggregates, geospatial datasets

Cloud & Infrastructure: AWS (EC2, RDS, S3, Lambda, Glue, EMR, CloudWatch), GCP (BigQuery, GKE, Cloud Functions), Terraform

Containers & Delivery: Docker, Kubernetes, Linux, containerised services, Cloudflare caching, GitHub, GitHub Actions, CI/CD

Data Pipelines: Dagster, Prefect, Airflow, dbt, Kafka, ETL/ELT, ingestion, validation and transformation, change data capture

Quality & Observability: pytest, property-based testing, data checks, error handling, data quality monitoring, DataDog

Storage & Warehousing: Snowflake, Redshift, Parquet, Zarr, S3, lakehouse

ML & AI: PyTorch, HF Transformers, TensorFlow/Keras, vision models, DSPy, Claude Agent SDK, Weights & Biases

Education

MSc Computational Modelling

University of Southampton, Southampton, UK, 2014 – 2015

PGCE Mathematics and Qualified Teacher Status

Manchester Metropolitan University, Manchester, UK, 2012 – 2014

BSc Mathematics, First Class Honours

Manchester Metropolitan University, Manchester, UK, 2009 – 2012